

LECTURE 1

Engineering Ethics

What it means, why engineers need it, and how to recognise it when you meet it at work.

Md. Owaliur Rahman

BSc in CSE, BUET

Lecturer, Presidency University

OPENING SCENARIO

The bridge

THE JOB

You are on the design team for a new road bridge. Construction starts in eight weeks.

WHAT YOU FOUND

Under one unusual but realistic load, a jam of heavy trucks in a storm, the safety margin drops below the code minimum.

THE FIX

Redesigning two piers. About six months of delay and a large amount of money.

So far nobody has lied, broken a rule, or done anything wrong.

YOU TAKE IT TO YOUR MANAGER

"The probability of failure is very small. We cannot stop the project over this."

He is not a villain either. He is responsible for a deadline, a budget, and the jobs of forty people.

DISCUSSION

What would you do?

Accept the answer and move on.

Refuse, and take it over his head.

Something in between.
What, exactly?

Five things pulling at you

Safety

A small chance is still a chance, and someone drives over it.

Cost

The fix is real money, and it comes out of a real budget.

Deadline

Six months late also has costs for people you never meet.

Your job

You are junior. Your manager writes your review.

The public

They never see the drawing. They trust it anyway.

Every one of these is a legitimate concern. None of them is greedy or stupid. That is what makes the decision hard.

Every option has a cost

If you stay quiet

- The project stays on schedule.
- The bridge will very probably be fine.
- If it is not, people are hurt, and you knew.

If you push back

- The project may slip by six months.
- Your manager will not thank you.
- You might be wrong, and the delay still costs.

There is no option here that costs nothing. Anyone who tells you otherwise has not understood the problem.

This is an engineering ethics problem.

Ethics is not about being a nice person. It is about deciding responsibly when real and legitimate interests pull in different directions.

Where we are going today

1

What ethics actually means

2

Ethics, morality and law

3

Why engineers in particular need it

4

How to spot an ethical issue

5

Who an engineer answers to

6

Back to the bridge, in groups

No theories today. No new vocabulary to memorise. Understanding first.

PART ONE

What is ethics?

Plain language only.

If you need a dictionary for it, I have explained it badly.

Three different questions

Can I do it?

Is it technically possible? Your degree answers this one.

Am I allowed to?

Does the law or the contract permit it? Lawyers answer this one.

Should I do it?

Is it the responsible thing to do? Nobody else answers this one for you.

The third question is ethics. Notice that you can answer yes to the first two and still be stuck on it.

DEFINITION

Ethics is the set of principles that help us decide what we ought to do.

AN EXAMPLE, RIGHT AWAY

Your app works. It also switches the microphone on while it is in the background. No law stops you. No test fails. Nobody asked the user.

THE ETHICAL QUESTION

Not can you ship it. Not are you allowed to ship it. Should you ship it that way?

Two questions, one job

FOUR YEARS OF ENGINEERING SCHOOL

Can we build it?

THIS COURSE

**Should we build it
this way?**

The second question does not replace the first. It is added to it. You need both to be a competent engineer.

PART TWO

Ethics, morality and law

Three words that get used as if they were the same word. They are not.

Three words people mix up

Morality

What you personally, or your family and community, believe is right and wrong.

I was raised to think lying is shameful.

Ethics

A reasoned way of deciding what ought to be done, shared inside a profession or a society.

An engineer does not hide a test result, whatever she was raised to think.

Law

Rules written and enforced by the state. You can be punished for breaking them.

Falsifying a safety certificate is a crime.

Side by side

	Morality	Ethics	Law
Main idea	Personal and social beliefs about right and wrong	Principles for making responsible decisions	Rules that can be enforced against you
Where it comes from	Family, culture, religion, upbringing	Reasoning, professional standards, society	Government and the courts
Who enforces it	Your own conscience, the people around you	Your profession, colleagues, employers	Police, regulators, the legal system

Copy this table. It is the only thing today worth writing down word for word.

Legal is not the same as right

WRONG AND ILLEGAL TOGETHER

An engineer signs off on a load test that was never carried out. That is dishonest, it endangers people, and it is also a crime.

LEGAL, AND STILL WRONG

A machine ships with the guard the standard requires. Your team knows a known pinch point sits just outside what the standard covers. Operators will reach into it. Nothing in the law says you must fix it.

Law and ethics overlap heavily. Most things that are seriously unethical are also illegal. The law is a floor, not a ceiling.

PART THREE

Why engineering needs ethics

Every profession has ethics. Engineering needs it for a specific reason.

Engineers have power

Not authority. Reach. What you design touches people who never asked for it and cannot inspect it.



Public safety



Buildings and
bridges



Electricity



Medical devices



Software



Personal data



Money systems



Communication



AI systems



The environment

One small engineering decision can reach thousands of people. Sometimes millions.

Three ordinary decisions

A software engineer decides not to report a security hole before release. The release date is next week.

Who is exposed? Every user of the product.

A civil engineer approves a cheaper grade of material. It is within tolerance on paper.

Who is exposed? Everyone who uses the structure for fifty years.

A data engineer logs each user location every thirty seconds, in case it is useful later.

Who is exposed? Anyone whose movements are now in a database.

None of these takes more than an afternoon to decide. None of them feels dramatic at the time.

DISCUSSION

Who pays the price if the engineer gets it wrong?

An engineering solution can be technically correct and still be ethically wrong.

Passing every test tells you the thing works. It does not tell you the thing should exist in that form.

The app that works perfectly

A team ships a photo app. It never crashes, it loads fast, it passes every test. In the background it uploads the user's contact list to the company's servers. Users are never told.

1

Is the software technically successful?

2

Is the decision ethically acceptable?

3

Why are these two different questions?

The machine that meets spec

A packaging machine hits every performance number in the contract. To hold that speed, the operator has to reach past an open guard to clear a jam. That happens about twenty times a shift.

TECHNICAL QUESTION

Does it meet the specification? Yes, on every line.

HUMAN QUESTION

Does it put a person at risk that a better design would have removed? Also yes.

Engineering judgement always has both sides. The specification only covers one of them.

PART FOUR

Ten signals that a decision needs a second look

Safety

Risk of harm

Honesty

Privacy

Fairness

Duties that
clash

Conflict of
interest

The
environment

Public welfare

Professional
duty

If one of these words applies to the decision in front of you, the decision has an ethical side. That is all this list is for.

THE RED FLAG TEST

Could my decision unfairly harm someone, deceive someone, expose someone to unnecessary risk, or break a responsibility I have taken on?

If the answer is yes, or even maybe, stop and look at the ethical side before you carry on.

DISCUSSION · FIVE MINUTES

Red flag or not?

A

You round a measured value from 4.7 to 5.0 in a report, because it is within tolerance anyway.

B

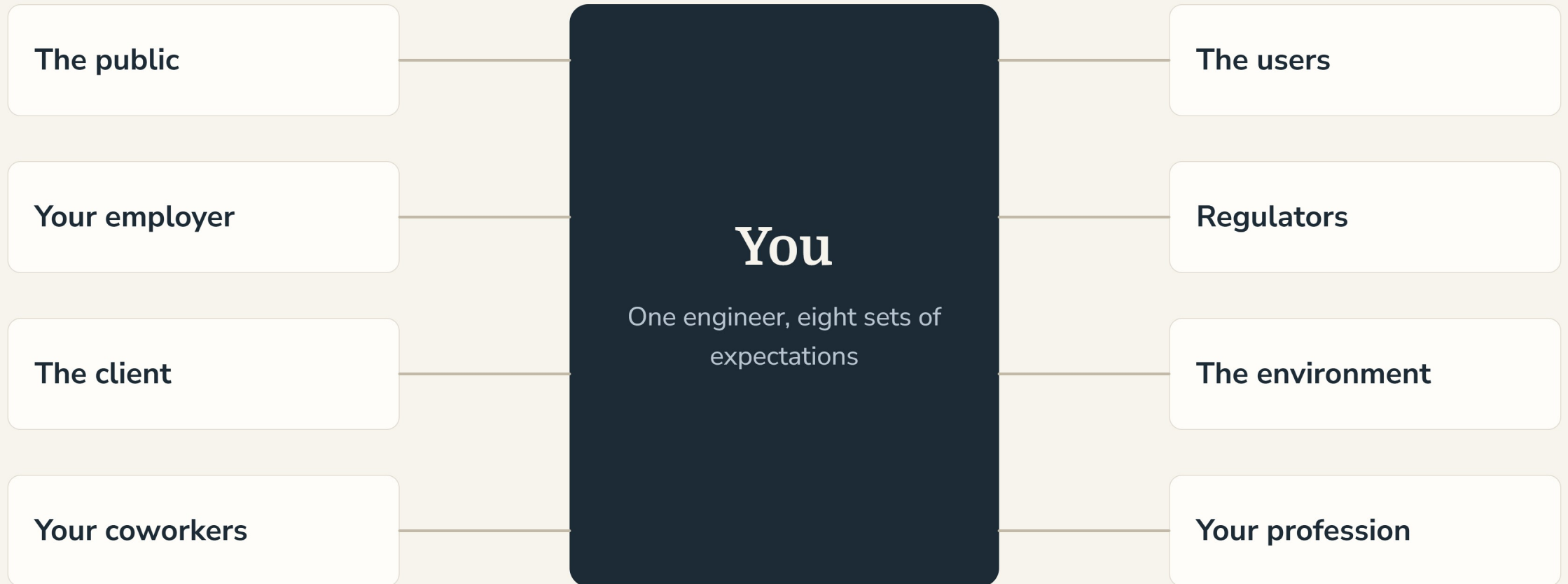
You recommend supplier B because it is cheapest. Your cousin owns supplier B.

C

You skip a test that has passed every time for two years, because the deadline is tomorrow.

For each one: is there a red flag? Which word from the last slide fits it?

An engineer answers to many people



When they want different things

YOUR EMPLOYER

**The cheapest
solution**

YOUR CLIENT

**The fastest
solution**

THE PUBLIC

**The safest
solution**

Whose interests should an engineer weigh, and in what order? We will not settle that today.

Your responsibility does not stop at your employer's door.

The people who drive over the bridge never signed your contract and never met you. They depend on your judgement anyway.

CASE STUDY

Back to the bridge

Same problem as an hour ago. You now have words for what is going on in it.

The bridge, in full

- 1** The load case is unusual but realistic: heavy trucks stopped on the span during a storm.
- 2** Your calculation shows the safety margin falling below the code minimum in that case.
- 3** One senior colleague agrees with you. Another says your load assumption is too pessimistic.
- 4** The fix is a redesign of two piers: roughly six months and a large amount of money.
- 5** The contract carries a late penalty, and your manager's bonus depends on the delivery date.
- 6** Nothing has been built yet. Everything is still on paper.

WORK

Six questions

1

What is the technical problem?

2

What is the ethical problem?

3

Who are the stakeholders?

4

What could go wrong, and how badly?

5

What responsibilities does the engineer carry?

6

Which interests are pulling against each other?

TRANSITION

You have not been given a method yet

TODAY

You can now see that there is an ethical problem, and say what it is made of.

NEXT CLASSES

A systematic way to work through a case like this, step by step, and what professional codes actually require.

Seeing the problem is the harder half. Most people who get it wrong never saw it at all.

THREE QUESTIONS TO KEEP

Can we do it?

Should we do it?

Who is affected if we get it wrong?

Ask all three, in that order, for the rest of your career.

SUMMARY

Six things to remember

- 1** Ethics helps us decide what we ought to do, not what we can do.
- 2** Law and ethics are not the same thing. The law is a floor.
- 3** Engineering decisions reach people who never meet the engineer.
- 4** Technical competence alone is not enough to make a decision right.
- 5** Engineers are responsible to the people their work affects.
- 6** Ethical problems appear when responsibilities and interests conflict.

Number six is the definition of a hard case. If nothing is in conflict, you do not need this course.

REVISION

Questions from this lecture

Thirteen questions in exam form. Everything here was covered today.

Foundations, law and ethics

PART A · FOUNDATIONS

- 1 Define engineering ethics.
- 2 Distinguish between the questions "Can I do it?" and "Should I do it?", with reference to engineering practice.
- 3 Differentiate between ethics, morality and law.

PART B · LAW AND ETHICS

- 4 Explain the statement: "The law is a floor, not a ceiling."
- 5 Can an act be legal and yet unethical? Justify your answer with a suitable engineering example.

Why engineering needs ethics

PART C · WHY ETHICS IS REQUIRED

- 6 Why is ethics important in the practice of engineering?
- 7 Why is technical competence alone not sufficient for an engineer? Support your answer with an example.

PART D · RECOGNISING AN ETHICAL ISSUE

- 8 What characteristics indicate that an engineering decision carries an ethical dimension?
- 9 Explain the red flag test used to identify an ethical issue.

Responsibilities and conflicts

PART E · WHO AN ENGINEER ANSWERS TO

- 10** Name the stakeholders an engineer may have to consider in a professional decision.
- 11** Define conflict of interest and conflict of responsibilities, and explain how each can give rise to an ethical issue.
- 12** Explain how environmental concerns can turn an engineering decision into an ethical issue.
- 13** Explain the statement: "Your responsibility does not stop at your employer's door."

Answer in your own words. An answer copied from the slides earns the marks for the slides, not for you.